Build a Paper Prototype

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 31: a complete decision rule includes a fallback.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Build a runnable paper prototype with explicit instructions and visible state. Try the recall before opening notes.

Your learning target

I can build a runnable paper **prototype** with explicit **instruction**s and visible state.

I can record what actually happens without silently repairing the **instruction**s.

I can use a **trace** to distinguish an **instruction** defect from an execution mistake.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

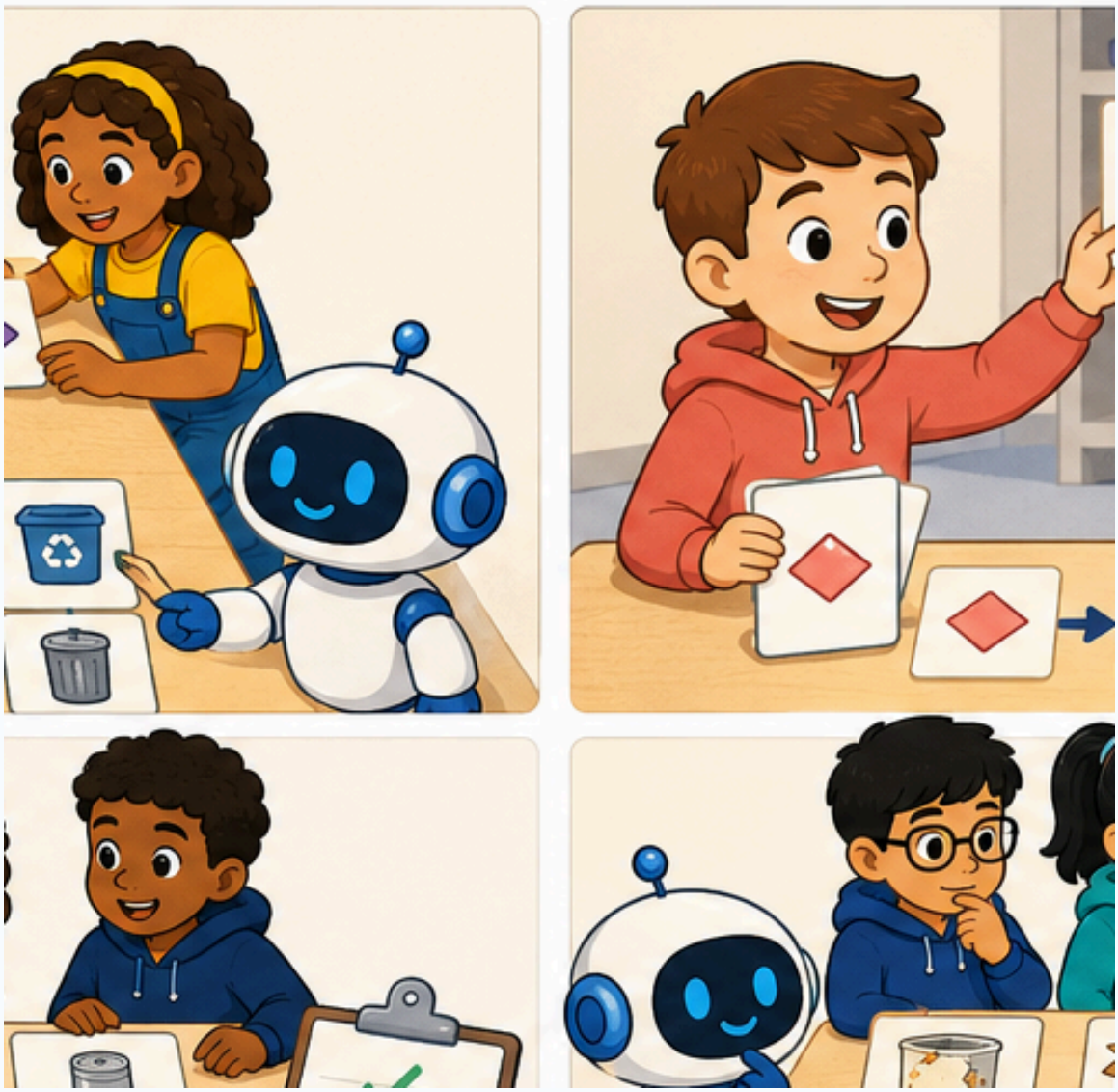
Without opening the old lesson, explain one key idea from Design the Decision **Process**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompts Are Design **Instruction**s. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A **prototype** tests the idea before polish or technology.



Your target

LEARNING TARGET Assign three people as **input** reader, rule follower and **output** recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes.

What does this **prototype** demonstrate?

Learner name:

Readiness for the connected explanation

When one candidate fails, what must happen next, and when is no match justified?

This week's end goal

By the end, I can build and operate a paper **prototype** while recording actual **trace**s.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Build a Paper Prototype case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY One student presents an item card while another follows the written sorter **process** exactly and returns the documented **output**.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

This week you will turn the decision **process** into a paper **prototype**. One partner operates the **instruction**s and another records the steps.

A working **prototype** needs more than a polished drawing: someone else must be able to follow it without asking its author to supply missing rules. Keep the version you ran so the **trace** can be checked against it.

Copy this catalogue in order onto four cards: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These are fictional paper-model values with materials assumed available. Label the catalogue B version 1.

Keep the Week 30 contract: **input** is a whole number of available minutes from 0 to 30, and Quiet required Yes or No.

An activity fits when its time is no more than the available minutes AND either quiet is not required or the activity is quiet. No means either kind is allowed. Suggest the first fit with a reason; otherwise report no match.

Missing or invalid **input** needs clarification. Suggestions do not book materials or publish anything.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Build a Paper Prototype. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



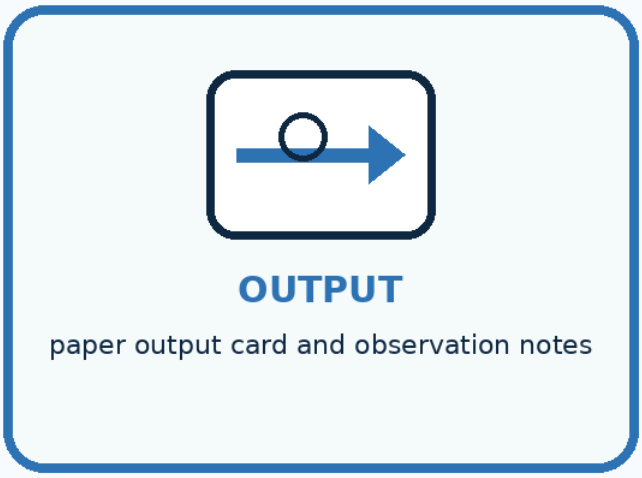
INPUT

paper item card



PROCESS

partner executes the process without improvising



INPUT paper item card

PROCESS partner executes the **process** without improvising

OUTPUT paper **output** card and **observation** notes

Key idea

CORE IDEA A **prototype** tests the idea before polish or technology.

How the mechanism works

UNDER THE HOOD A low-fidelity **prototype** externalizes assumptions cheaply.

Wizard-of-Oz **simulation** is valid when the human **operator** follows the specified **process** and does not secretly add judgment unavailable to the proposed system.

Draw a fresh example with all three stages:

A closer explanation

A paper run that can be checked

PREPARE THE RUN	OPERATE ONE STEP	KEEP THE EVIDENCE
Request and version labels	Validate the two fields	Record pass or clarification
Marker at current row	Check time AND quiet	Record checks and next step
First fit or end of list	Suggest or report no match	Record output and stop

The request stays fixed while the marker advances. Keep the instruction version and trace together.

Make a request card with Available minutes and Quiet required fields, a marker for the current catalogue row, and a **trace** sheet. Head the sheet Run ID, **prototype** version, catalogue version and request.

Below it use columns Step, current row, time check, quiet check and next action or **output**. Use not checked when validation stops the search.

Write **instruction** cards P1 version 1. Card 1: validate both fields; if invalid, name the field needing correction and stop; otherwise put the marker at B1.

Card 2: check the marked row against both compatibility conditions; record both checks. If both pass, suggest that row with the reason and stop.

Card 3: if either fails and another row remains, advance exactly one row and return to Card 2; if no row remains, report no match and stop. Begin each new request at Card 1 with a cleared **output**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

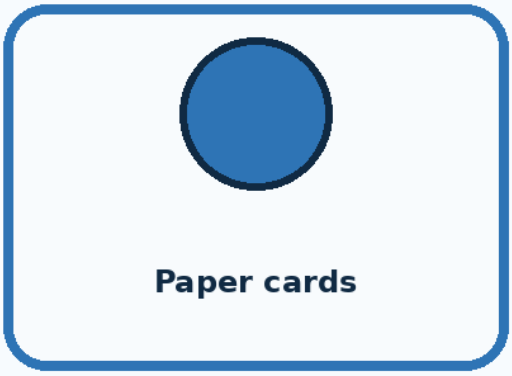
Compare the cases

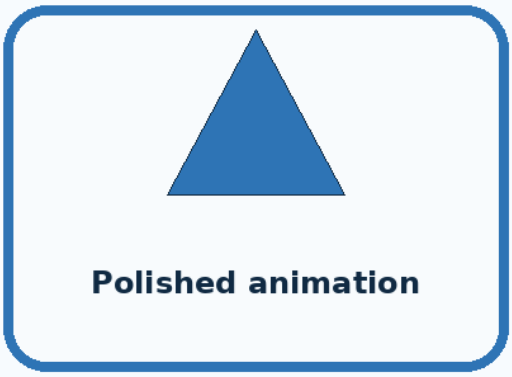
> Compare the supplied cases for Build a Paper Prototype. Identify the detail that changes the conclusion.

PICTURE GALLERY

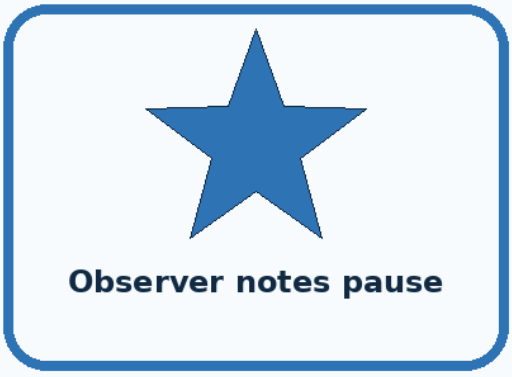
Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Paper cards Clue: fast to revise. Meaning: good for logic test.

The case to investigate

CASE 2 — Polished animation Clue: looks complete. Meaning: may hide weak **process**.

The case to investigate

CASE 3 — **Operator** improvises Clue: adds secret knowledge. Meaning: invalid **simulation**.

The case to investigate

CASE 4 — Observer notes pause Clue: reveals confusion. Meaning: useful **evidence**.

Keep this boundary clear

BOUNDARY A helpful **operator** can accidentally make a weak design look successful by using unstated human knowledge.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Insert “if capacity is below count, continue” before accepting a bin.

Predicted result

Prediction: The revised **trace** rejects X and accepts Y for 4.

Why the result follows

Reason: 3<4 is true; 7<4 is false, so Y fits.

STUDENT LESSON · 5 OF 18

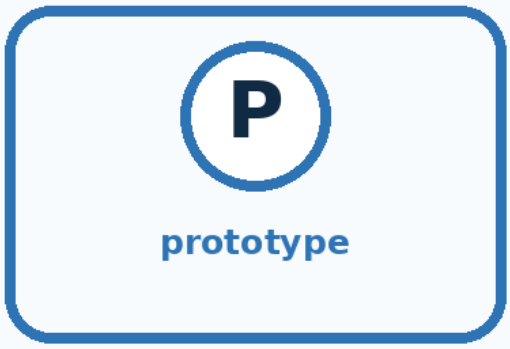
Words that unlock the idea

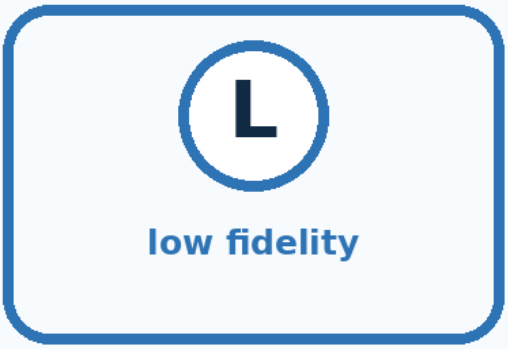
> Use the vocabulary of Build a Paper Prototype to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
prototype	early representation used to test an idea	
low fidelity	simple version focused on function	

Word	Meaning	My example
simulation	acting out system behavior	
operator	person executing the prototype process	

Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR A **prototype** must look like the final product.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Instruction	A precise action or decision for the operator to follow
Trace	A record of the steps and state during one run
Expected result	The result required by the agreed specification
Observed result	What actually happened when this version was run

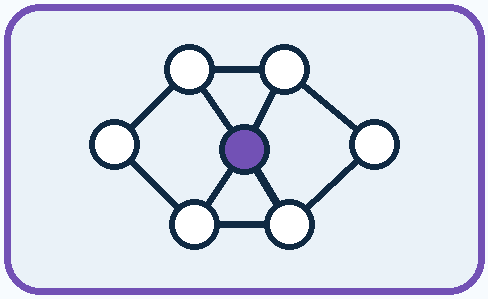
Follow a worked example

> Follow the worked trace for Build a Paper Prototype, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE One student presents an item card while another follows the written sorter **process** exactly and returns the documented **output**.

Step 1

STEP 1 — OBSERVE paper item card

Step 2

STEP 2 — **TRACE** partner executes the **process** without improvising

Step 3

STEP 3 — CONCLUDE paper **output** card and **observation** notes

Model answer

MODEL REASONING The **operator** must use only documented **input**s and steps. Any question, pause, or improvisation reveals a missing rule, unclear label, or unsupported **input**. Revise the paper flow before adding technology.

A claim to avoid

DO NOT OVERCLAIM A helpful **operator** can accidentally make a weak design look successful by using unstated human knowledge.

What new **evidence** could change this answer?

A closer explanation

Worked explanation · Build a Paper Prototype

Before operating P1, use the full manual checklist to write the **expected result** for 9 minutes, Quiet required No. Then run the cards literally. Do not replace a surprising result with the expected one.

Recording disagreement gives you something to investigate; hiding it removes useful **evidence**.

Trace step for 9 minutes No	Checks recorded	Next action or output
1 Validate	Whole minutes in range; No accepted	Place marker at B1

Trace step for 9 minutes No	Checks recorded	Next action or output
2 Check B1	10 is greater than 9; quiet unrestricted	Move to B2
3 Check B2	8 is no more than 9; quiet unrestricted	Suggest B2 with reason; stop

Worked explanation · Build a Paper Prototype

This **trace** agrees with the expected B2 result. It shows that this run followed the **instruction**s. One matching result does not prove that every request will work or that the **prototype** is faster than the manual checklist.

This is a system of **written rule**s, not a model trained on examples.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Build a Paper Prototype

A fictional team turns a storage-bin specification into movable **instruction** cards.

The facts we will use

Supplied facts

Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

Supplied facts

For request 4, write baseline Y before operating cards. **Prototype** cards mistakenly stop with X immediately.

AI ACADEMY / AI-1/32

Build a Paper Prototype

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Expected result

Baseline checks X then Y and expects Y for count 4.

2. Operate literally

Follow the current prototype cards without secretly repairing them.

3. Actual trace

Prototype stops at X; it never performs the capacity comparison.

4. Record mismatch

Expected Y, actual X: locate the missing check.

CHANGE ONE THING

Insert "if capacity is below count, continue" before accepting a bin.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Expected result	Baseline checks X then Y and expects Y for count 4.
2. Operate literally	Follow the current prototype cards without secretly repairing them.
3. Actual trace	Prototype stops at X; it never performs the capacity comparison.
4. Record mismatch	Expected Y, actual X: locate the missing check.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

What this example does not establish

Keep the stated limit

One repaired paper case does not establish usability, correctness on every **input** or performance of a later software implementation.

Keep the stated limit

Read the labelled picture: What should appear in the test log before and after the repair?

Keep the stated limit

Change one thing: Insert “if capacity is below count, continue” before accepting a bin. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Build a Paper Prototype claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Paper cards	fast to revise	good for logic test

Case	Evidence	What it shows
Polished animation	looks complete	may hide weak process
Operator improvises	adds secret knowledge	invalid simulation
Observer notes pause	reveals confusion	useful evidence

Repair two tempting claims

Challenge claim A

CLAIM A A **prototype** must look like the final product.

Repair:

Challenge claim B

CLAIM B If a person can perform the task, the written **process** is complete.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

Swap roles and give your partner the request card, catalogue, **instruction** cards and blank **trace** sheet. Let them operate without extra hints from the author.

Record any point where they cannot decide what to do from the written **instruction**s. Label and save changes, then repeat the affected run.

Keep development cases separate from any cases your teacher has reserved for a later evaluation.

Check the claim against the facts

Your handover should let someone repeat a run using the same **input**s and versions and check why it stopped. Keep all results, including disagreements. Next week you will organise several routine cases into a test record.

A neat interface is useful, but correctness depends on validation, both conditions, the first-match rule and a clear ending.

A saved trace says “P1 version 1; 11 minutes No; started at B2; suggested B2.” Explain the error using the written cards. State what evidence to keep and what should happen in a new correct run.

STUDENT LESSON · 8 OF 18

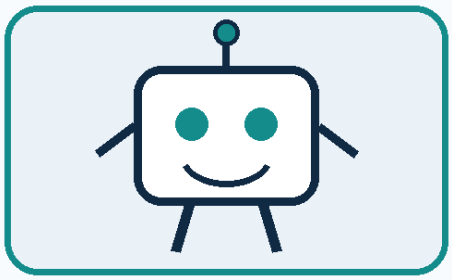
Practise together

Practise Build a Paper Prototype with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A **prototype** tests the idea before polish or technology.

Use a fresh example to explain **prototype**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: If a person can perform the task, the written **process** is complete.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Suppose the **operator** starts a fresh 11-minute No request at B2 because the marker was left there. They return B2. The required first fit is B1, and Card 1 explicitly says to restart at B1.

The **trace** therefore reveals an execution mistake. If the written **instruction**s had omitted the reset, that would instead be an **instruction** defect. Inspect the saved version before deciding.

Preserve the first **trace**. For an execution mistake, rerun the same written version correctly and label the new run. For an **instruction** defect, save a corrected version with a new label, state what changed, and rerun.

Never rewrite an old **observed result** to make a version appear successful.

> **Run P1 for 8 minutes with Quiet required Yes. Record every row, both checks and the final output. Then consider a blank Quiet required field: which card stops that run and how many candidates should be checked?**

Practise with less help: Run and record the first match

Use workbook W2 for the connected example “Building and tracing a paper **prototype**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your **trace** to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Building and tracing a paper **prototype**. Proposed explanation: “A correct-looking answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> What should appear in the test log before and after the repair?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

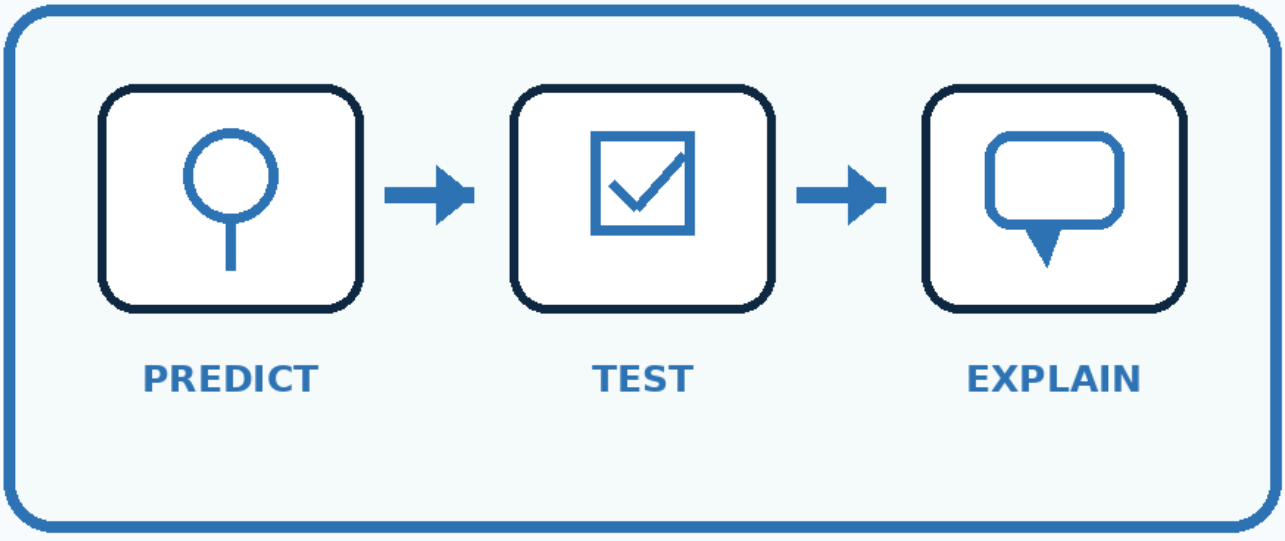
Run the investigation

> Investigate Build a Paper Prototype. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Partner-as-Computer Prototype

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Run the sorter with a partner acting as the computer.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only **trace**d the procedure.

STUDENT LESSON · 10 OF 18

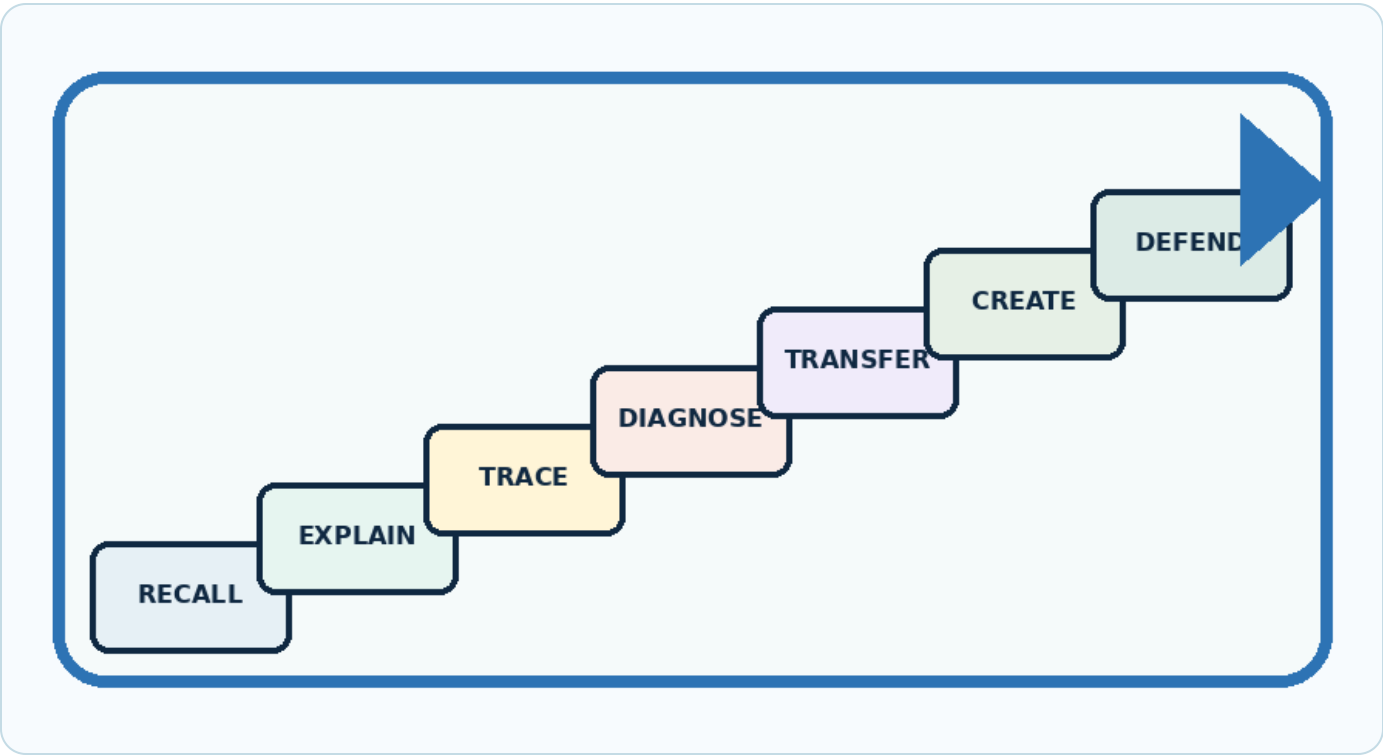
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Build a Paper Prototype. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **prototype**.

EXPLAIN A **prototype** tests the idea before polish or technology.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: A **prototype** must look like the final product.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

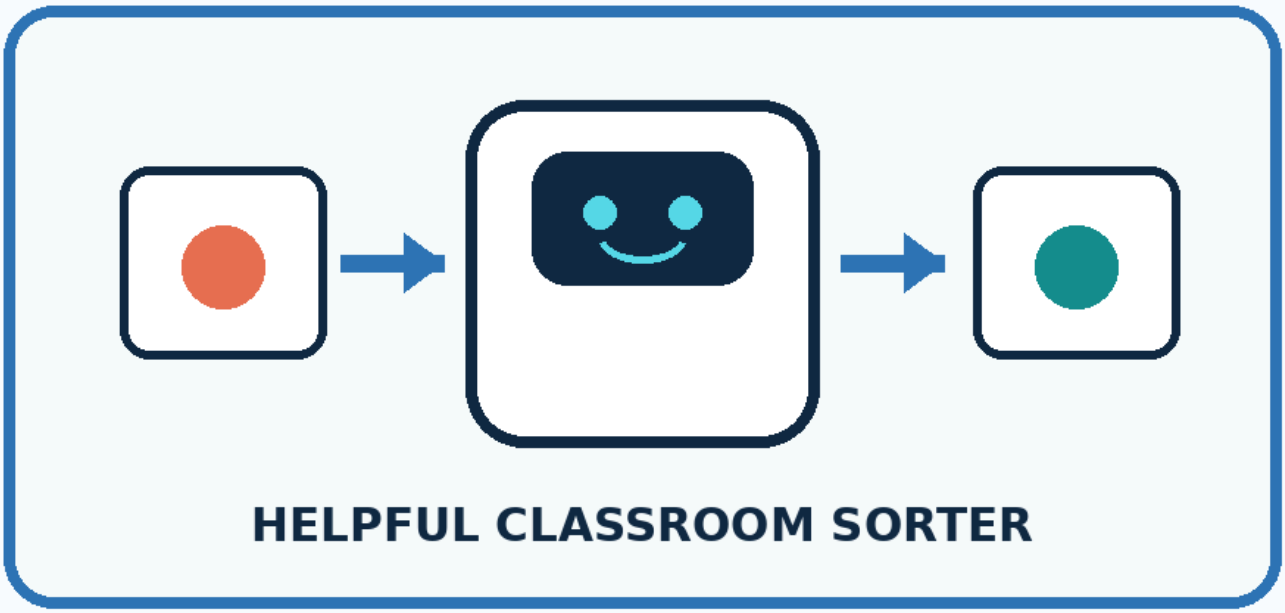
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Build the complete paper sorter: **input** cards, **process** guide, **output** cards, not-sure card, and test log.

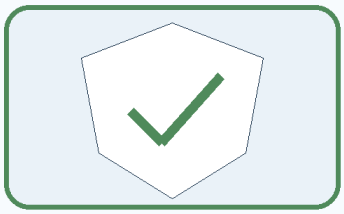
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN A helpful **operator** can accidentally make a weak design look successful by using unstated human knowledge.

My project decision:

Evidence for your project

Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s.

STUDENT LESSON · 12 OF 18

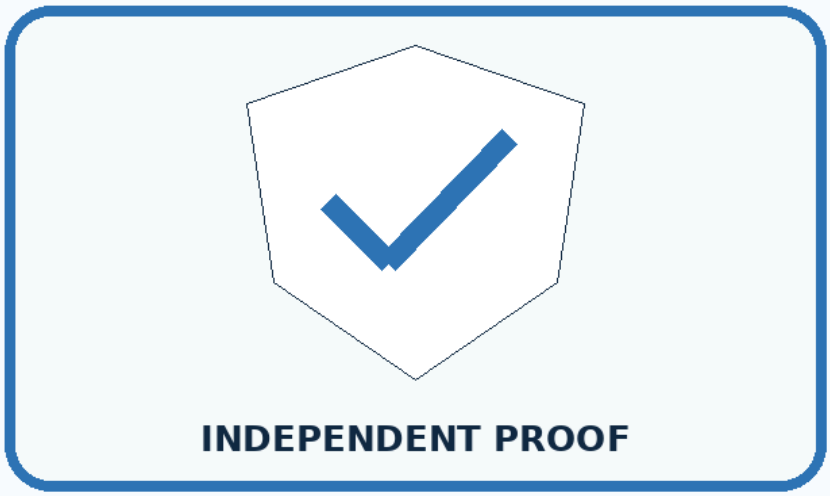
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Run the sorter with a partner acting as the computer and record where

the written **process** required guessing.

- ☐ Name or draw the fresh case.
- ☐ **Trace input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK The **operator** must use only documented **input**s and steps. Any question, pause, or improvisation reveals a missing rule, unclear label, or unsupported **input**. Revise the paper flow before adding technology.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Test Normal Cases.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Build a Paper Prototype case for supported practice. Repeated examples do not become fresh assessments.

Week 32 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A paper **prototype** has **input** cards, decision cards and paper, plastic and review trays.

The question

> How can a learner simulate it?

Worked explanation

Read **input**, follow written decisions and place a token in a tray. Record the **trace**. Human **simulation** does not establish model **training**.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Act out a paper prototype

Evidence or stage	Value or result
Input card	Observe material

Evidence or stage	Value or result
Decision cards	Follow written checks
Token	Place in paper, plastic or review tray
Trace	Record each decision

A paper **prototype** has **input** cards, decision cards and paper, plastic and review trays.

> How can a learner simulate it?

Read **input**, follow written decisions and place a token in a tray. Record the **trace**. Human **simulation** does not establish model **training**.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Build a new route through a small catalogue

Independently make and **trace instruction** cards for this paper equipment catalogue: E1 first, capacity 3 people; E2 second, capacity 6. Valid **input** is a whole group size from 1 to 8.

Suggest the first item whose capacity is at least the group size; no booking occurs. Include invalid **input** and no match. Write **expected result**s, then operate your cards for groups of 5 and 7.

Record both runs and explain every stopping point.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 32 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Your paper **prototype** shows an **input** card, a decision card and an **output** card. Explain what a person must simulate and what the **prototype** cannot prove.

My answer, with a **trace**, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and **expected result**

Write or draw your reasoning and evidence.

Connect to your project

Build the complete paper sorter: **input** cards, **process** guide, **output** cards, not-sure card, and test log.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Assign three people as **input** reader, rule follower and **output** recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes. What does this **prototype** demonstrate?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Build a Paper Prototype, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional team turns a storage-bin specification into movable **instruction** cards.

By the end, I can	How I will show it
Build a runnable paper prototype with explicit instructions and visible state	Explain the deciding step, show a trace or test, and state one limit.
Record what actually happens without silently repairing the instructions	Explain the deciding step, show a trace or test, and state one limit.
Use a trace to distinguish an instruction defect from an execution mistake	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Build a Paper Prototype without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Build a runnable paper **prototype** with explicit **instruction**s and visible state

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional team turns a storage-bin specification into movable **instruction** cards.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Expected Y, actual X: locate the missing check.

1. Actual **trace**

2. Record mismatch

3. Operate literally

4. **Expected result**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Record mismatch Expected Y, actual X: locate the missing check.

A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Insert “if capacity is below count, continue” before accepting a bin. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The revised **trace** rejects X and accepts Y for 4.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The revised **trace** rejects X

> and accepts Y for 4. 3<4 is true; 7<4 is false, so Y fits.

Repair a missing building block

Review Design the Decision **Process** (ai-1/31). Goal: Design a decision **process** that covers valid, invalid and no-match requests.

Short practice: Where does the first original rejection occur, and why is K3 not originally selected?. Return to this week and explain what you changed.

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Review Prompts Are Design **Instruction**s (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

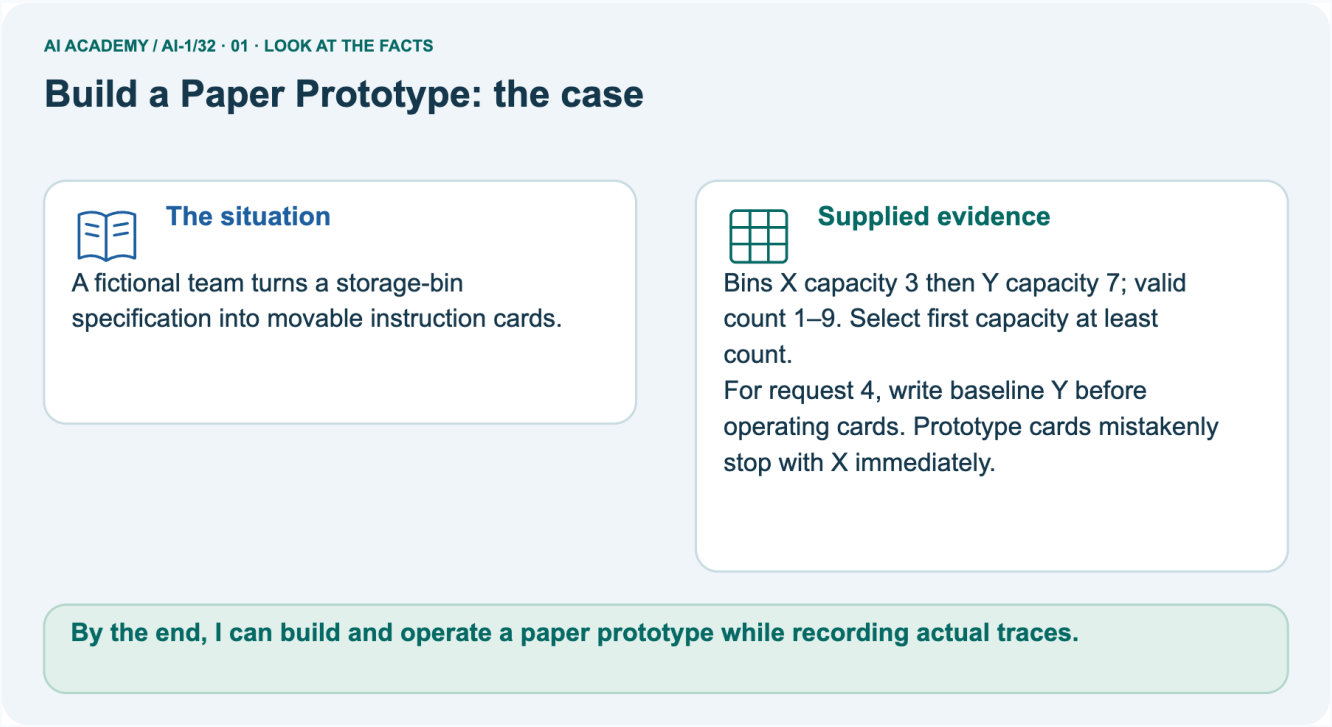
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Build a Paper Prototype graphic using its labels, deciding evidence and relevant limitation.

01. Build a Paper Prototype: the case



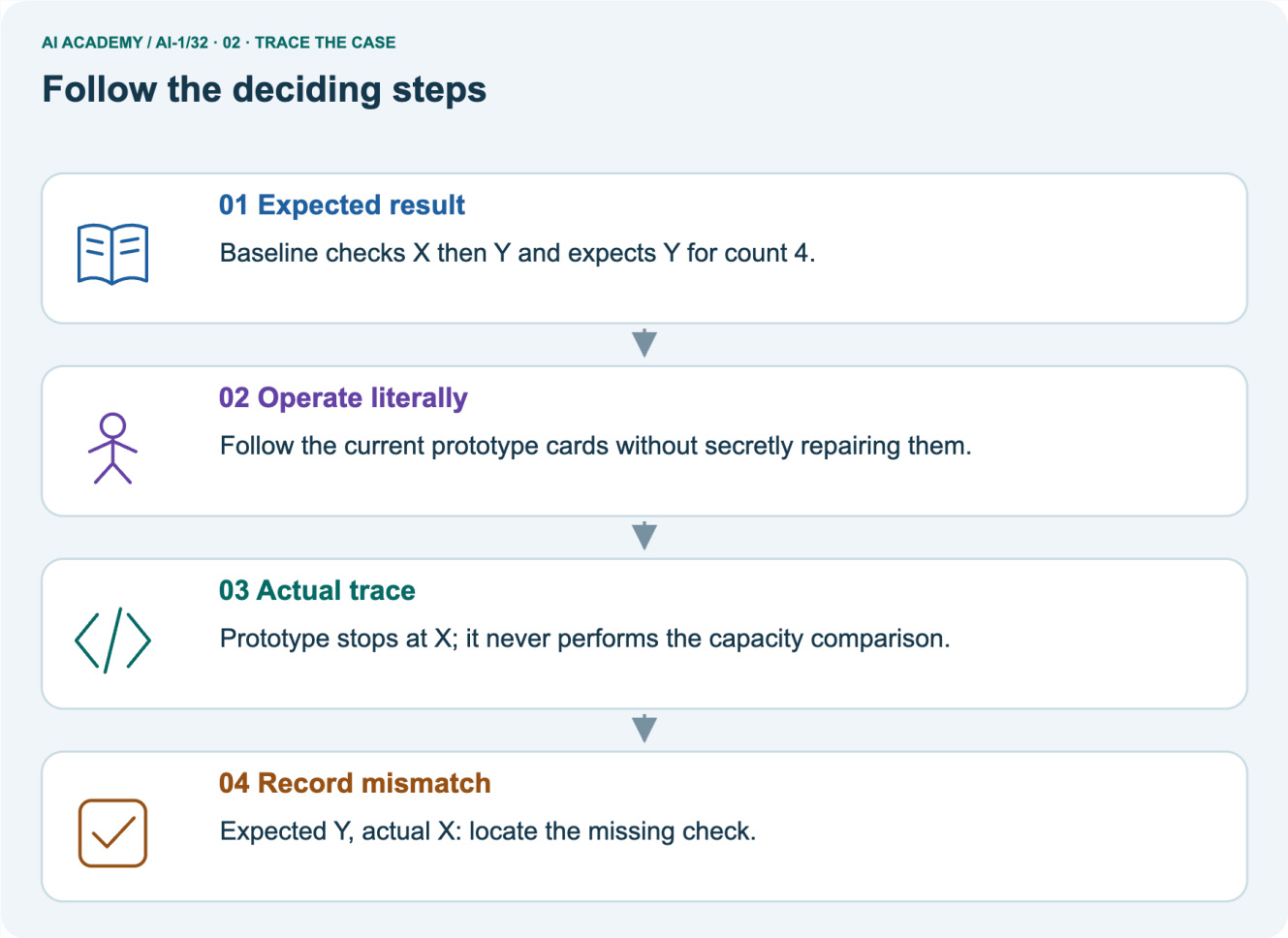
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional team turns a storage-bin specification into movable **instruction** cards. Supplied facts: Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

For request 4, write baseline Y before operating cards. **Prototype** cards mistakenly stop with X immediately.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

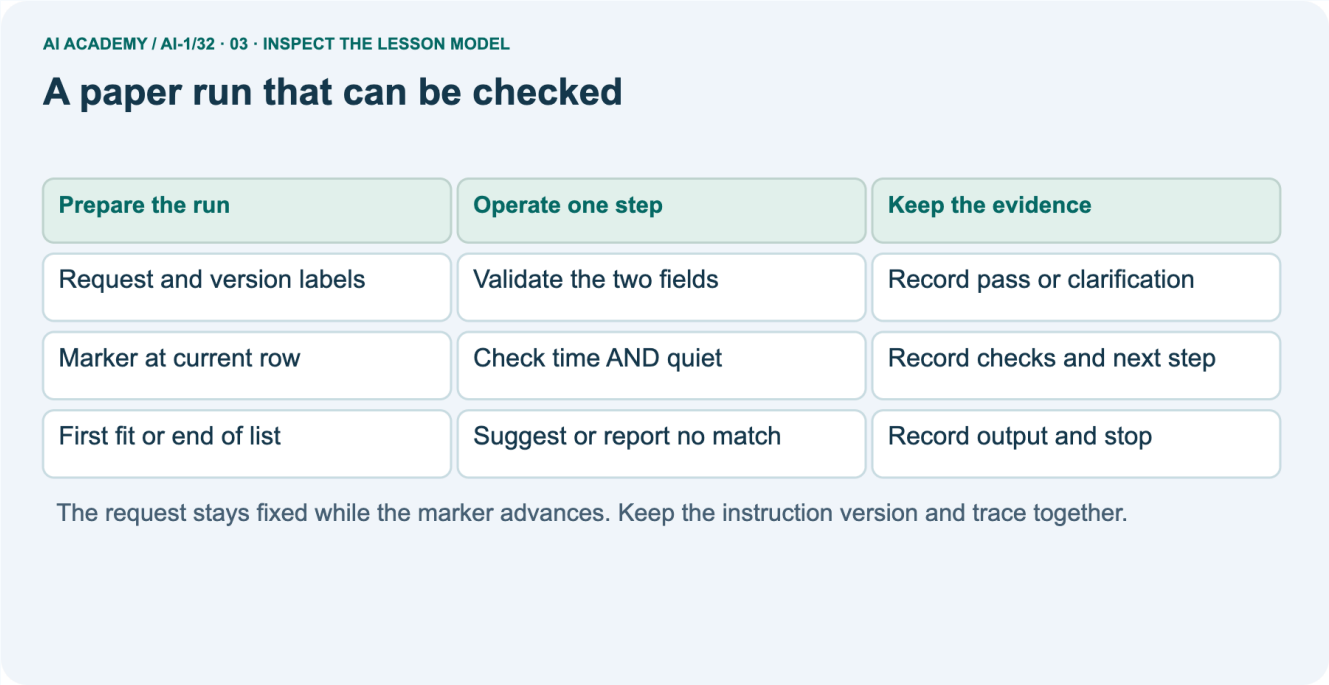


Follow the labelled picture with the matching explanation. Open the vector diagram

Expected result: Baseline checks X then Y and expects Y for count 4. → Operate literally: Follow the current **prototype** cards without secretly repairing them. → Actual **trace**: **Prototype** stops at X; it never performs the capacity comparison. → Record mismatch: Expected Y, actual X: locate the missing check.

> Point to each numbered stage and explain which recorded **input** it uses.

03. A paper run that can be checked



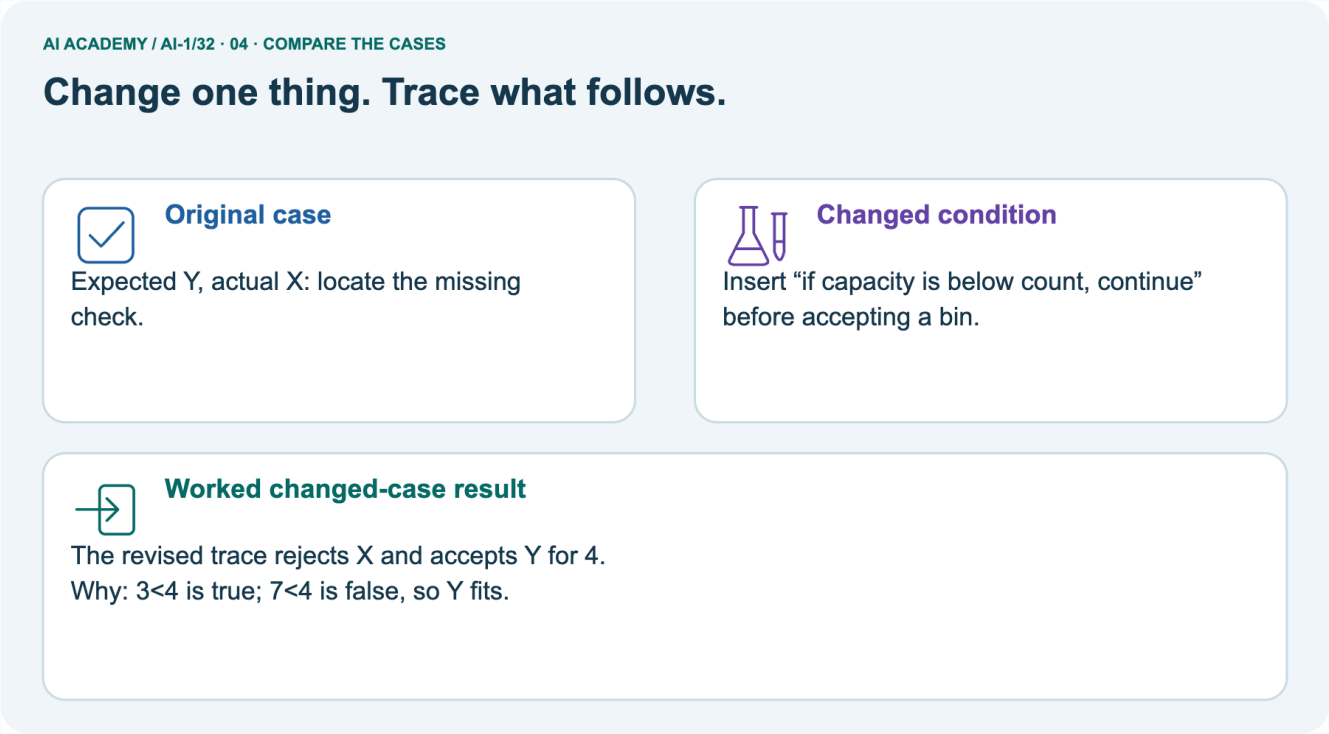
Follow the labelled picture with the matching explanation. Open the vector diagram

A paper run that can be checked. Request and version labels | Validate the two fields | Record pass or clarification. Marker at current row | Check time AND quiet | Record checks and next step.

First fit or end of list | Suggest or report no match | Record **output** and stop

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



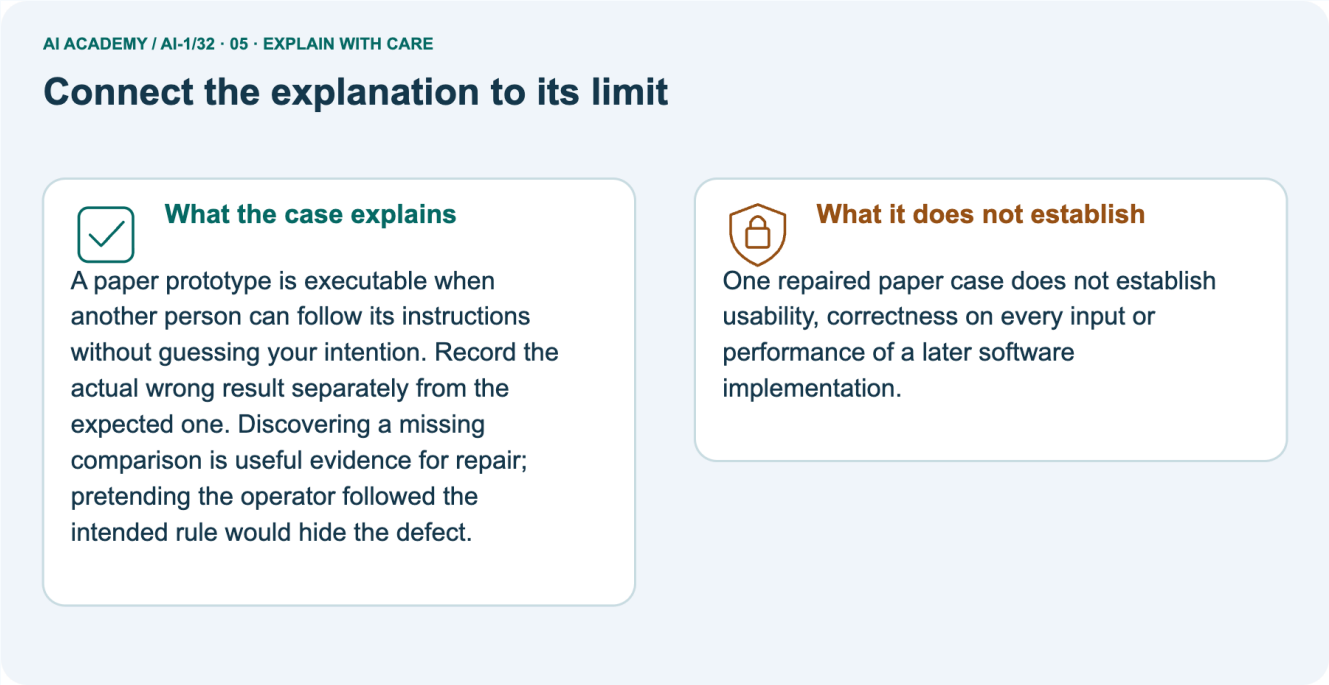
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Expected Y, actual X: locate the missing check. Changed condition: Insert “if capacity is below count, continue” before accepting a bin. **Expected result**: The revised **trace** rejects X and accepts Y for 4.

Reason: 3<4 is true; 7<4 is false, so Y fits.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

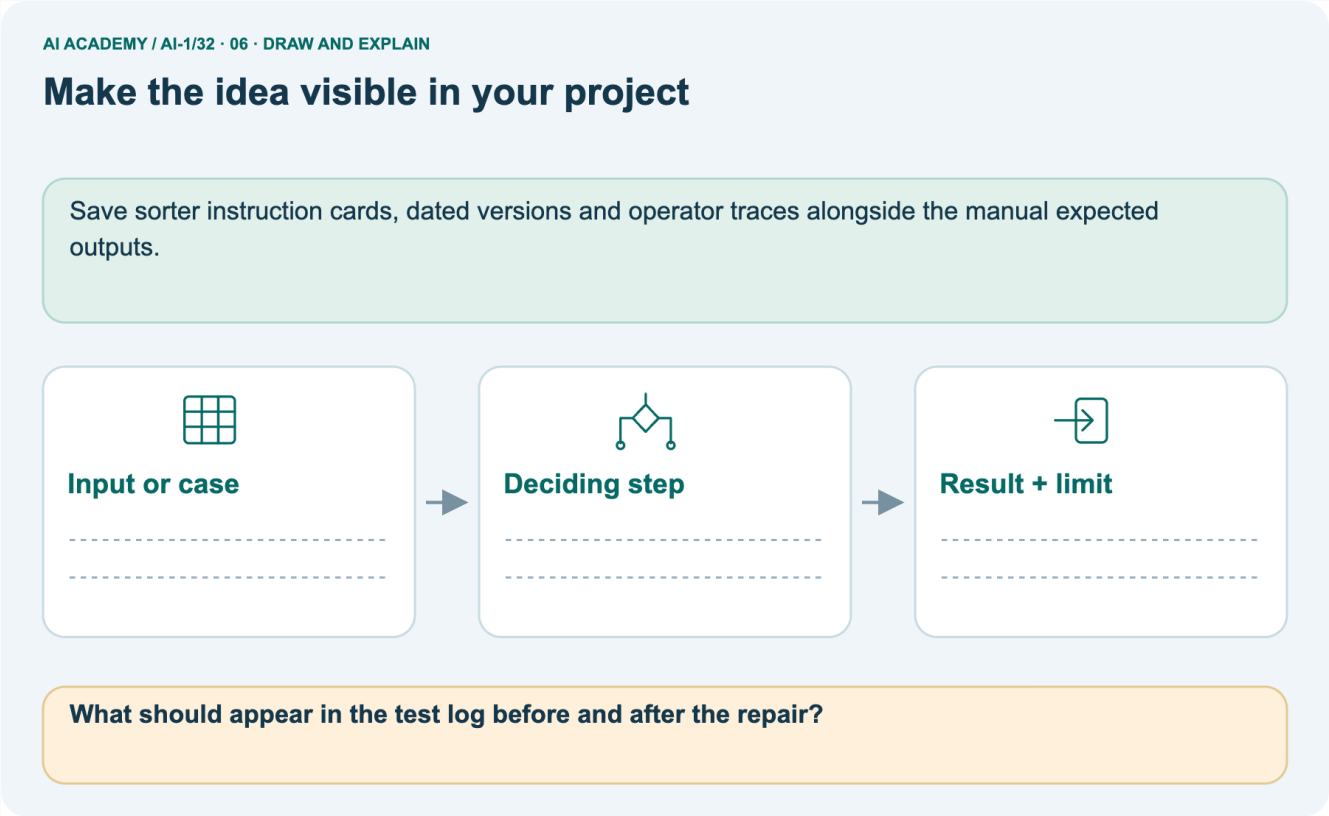
A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

Limit: One repaired paper case does not establish usability, correctness on every **input** or performance of a later software implementation.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s. Draw your **input**, deciding step, result and limit.

Supported question: What should appear in the test log before and after the repair?

Draw your project **trace** and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.